

HONGJING ZHU

Shenzhen Technology University, Guangdong, China
emiyarei33@gmail.com | (+86) 15107542067 | [33m4ximilian.github.io](https://github.com/33m4ximilian)

RESEARCH INTERESTS

Undergraduate student working on robotics, with experience focused on reinforcement learning for dexterous manipulation. I have independently modeled robotic tasks and carried them through simulation-based training to real-world deployment. My research interests include representation learning and commonsense reuse, memory and planning for long-horizon tasks, and adaptation through continued interaction. I am also interested in how robots can build on prior knowledge and experience to handle new tasks, and how deliberative planning and reactive control can work together. I remain open to interdisciplinary research and alternative approaches, and look forward to expanding my knowledge and learning new methods.

EDUCATION

Shenzhen Technology University

Sep 2024 – Present

B.Eng. Candidate, Unmanned Aircraft Systems Engineering | GPA: 3.0/4.0

INTERNSHIP EXPERIENCE

BrainCo — Dexterous Hand Algorithm Intern

Feb 2026 – Jun 2026

Open-source contribution: [RevoLab](#)

HORA/RMA In-Hand Rotation — Task Modeling, Training, and Deployment

- ★ Adapted HORA/RMA to ball and cylinder rotation on the Revo3 dexterous hand; built task environments and implemented two-stage teacher–student training using proprioceptive history.
- ★ Designed rewards for rotation speed, grasp stability, and joint constraints; applied domain randomization to object and actuator properties to account for physical variability.
- ★ Built a closed-loop PyTorch/ONNX deployment pipeline, aligning observation history and action interfaces between simulation and hardware; demonstrated the trained policy on hardware at the ICRA 2026 corporate exhibition booth.

Dynamic Handover — Dual-Arm Task Modeling and Policy Training

- ★ Reproduced dynamic throwing and catching for the RevoTron dual-arm platform; formulated a command-conditioned task for training a single policy across handover directions.
- ★ Implemented staged rewards for preparation, throwing, tracking, and catching, and integrated contact handling into the training environment.
- ★ Built parallel training and evaluation workflows; analyzed failed trajectories to refine contact handling and grasp stability, and evaluated policies across handover directions.

Robot Policy Training — Model Adaptation and Evaluation

- ★ Standardized image, robot-state, and action-sequence data in LeRobot; connected preprocessing, policy training, and offline evaluation through a shared workflow.
- ★ Adapted training and evaluation workflows for ACT, $\pi 0.5$, and GR00T N1.6, with baselines on PushT and ALOHA; checked temporal alignment and action outputs for data–policy compatibility.
- ★ Completed LoRA fine-tuning and multi-GPU training adaptation; resolved memory and distributed-training issues and implemented checkpoint-based training resumption.

RESEARCH MANUSCRIPTS

Human-Aware Physical Intelligence for Adaptive Human–Robot Collaboration: Mechanisms, Applications and Challenges

2026

[Hongjing Zhu*](#), [Haotong Luan*](#), et al. *Equal contribution / Co-first authors. Submitted to Smartbot; [under review](#).

Review on underactuated robotic hands and grippers — co-first author; in preparation for submission.

HONORS & AWARDS

2025 AI Robot Simulation Development Challenge — Undergraduate First Prize

Jan 2026

Lanqiao Cup Python Programming Competition — Provincial Second Prize

May 2025